

# AURORA artifact repository (GitHub)



github.com/SuviDeSilva94/AURORA

**SuviDeSilva94 / AURORA**

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AURORA — Dual-gated active inference for trustworthy self-healing on the Computing Continuum.

active-inference

bayesian-networks

edge-computing

self-healing

computing-continuum

## README.md

# AURORA: Uncertainty-aware resilience micro-agent

Reference implementation of the dual-gated active-inference framework evaluated in the companion thesis.

Reproduces all four-workload results:

- \* CCTV streaming (synthetic): 30,006 trials
- \* Motor monitoring (synthetic): 30,006 trials
- \* AI4I 2020 (real, UCI): 264 trials
- \* Steel Plates Faults (real, UCI): 621 trials

## Quick start

```
$ pip install -r requirements.txt
$ python experiments/run_comparison.py
```

## License

MIT

## Repository layout

```
AURORA/
  src/aif/
    bayesian_network.py
    two_stage_gate.py
    vfe_compute.py
    root_cause_analyzer.py
  experiments/
    run_comparison.py
    run_motor_comparison.py
    run_real_data_comparison.py
    run_steel_plates_comparison.py
  results/
  tests/
    test_aif.py
  requirements.txt
  LICENSE
```